For a satellite of mass M to move in a circular orbit, centripetal force must be acting on the satellite. This centripetal force is provided by the Earth's gravitational force.

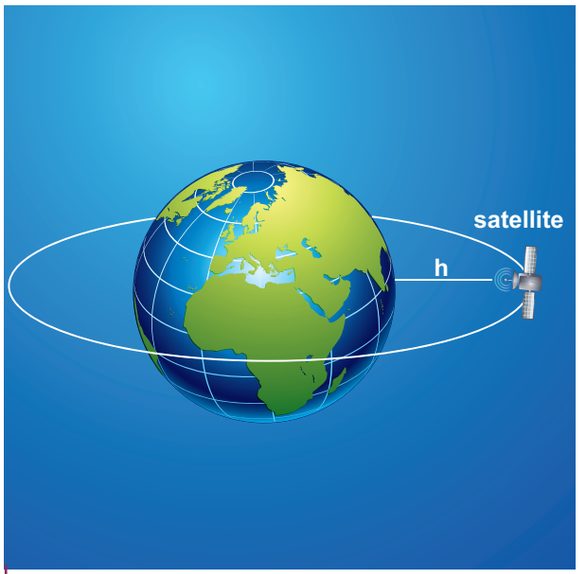


Figure 6.20 Satellite revolving around the Earth.

$$\frac{Mv^2}{(R_E+h)} = \frac{GMM_E}{(R_E+h)^2} \qquad (6.57)$$

$$v^2 = \frac{GM_E}{(R_E+h)}$$

$$v^2 = \sqrt{\frac{GM_E}{(R_E+h)}} \qquad (6.58)$$

As h increases, the speed of the satellite decreases.

Time period of the satellite:

The distance covered by the satellite during one rotation in its orbit is equal to $2\pi(R_E+h)$ and time taken for it is the time period, T. Then

$$Speed\ v = \frac{Distance\ travelled}{Time\ taken} = \frac{2\pi(R_E+h)}{T}$$

From equation (6.58)

$$\sqrt{\frac{GM_E}{(R_E+h)}} = \frac{2\pi(R_E+h)}{T} \qquad (6.59)$$

$$T = \frac{2\pi}{\sqrt{GM_E}}(R_E+h)^{3/2} \qquad (6.60)$$

Squaring both sides of the equation (6.60), we get

$$T^2 = \frac{4\pi^2}{GM_E}(R_E+h)^3$$

$$\frac{4\pi^2}{GM_E} = \text{constant say } c$$

$$T^2 = c(R_E+h)^3 \qquad (6.61)$$

Equation (6.61) implies that a satellite orbiting the Earth has the same relation between time and distance as that of Kepler's law of planetary motion. For a satellite orbiting near the surface of the Earth, h is negligible compared to the radius of the Earth R_E. Then,

$$T^2 = \frac{4\pi^2}{GM_E}R_E^3$$

$$T^2 = \frac{4\pi^2}{GM_E/R_E^2}R_E$$

$$T^2 = \frac{4\pi^2}{g}R_E$$

$$since\ GM_E/R_E^2 = g$$

$$T = 2\pi\sqrt{\frac{R_E}{g}} \qquad (6.62)$$

UNIT-6(XI-Physics_Vol-2).indd 24 17-08-2020 16:15:44